# Computational Identification of Collateral Sensitivity Flipper Mutations in *Burkholderia* via Lineage-Aware Genomic Analysis

Akhi C. Chappidi
Department of Computer Science
University of North Carolina at Charlotte
Email: achappid@charlotte.edu

Advisor: Dr. Way Sung
Department of Bioinformatics and Genomics
University of North Carolina at Charlotte
Email: wsung@charlotte.edu

***Abstract*—Chronic and recurrent infections caused by members of the *Burkholderia cepacia* complex (Bcc) present major clinical challenges for individuals with cystic fibrosis (CF). These opportunistic pathogens possess broad-spectrum antibiotic resistance and exhibit rapid adaptive evolution during treatment, often leading to multidrug resistance (MDR). One phenomenon with therapeutic potential is collateral sensitivity (CS), in which resistance to one antibiotic induces increased sensitivity to another. Reciprocal collateral sensitivity (RCS) pairs may enable alternating antibiotic therapies that suppress long-term resistance evolution. However, the genetic mechanisms underlying CS remain poorly defined, and experimental evolution studies generate large, complex lineage datasets that are difficult to analyze manually.**

**This study developed scalable Python pipelines to detect mutation "flipping" events—alleles that repeatedly appear and revert across evolutionary lineages as sensitivity profiles change. Whole-genome sequencing data from experimentally evolved *Burkholderia* lineages exposed to six clinically relevant antibiotics (CHL, CAZ, LVX, MEM, MIN, and SXT) were analyzed using parallel SNP and indel workflows. These pipelines performed input parsing, lineage-aware mutation comparison, allele-frequency validation, functional annotation, homopolymer analysis, and transition-metric scoring.**

**Across the indel workflow, 297 candidate flipper rows were identified and reduced to 14 high-confidence sites using read-frequency filtering. Several candidates reached 100% allele frequency and mapped to genes involved in efflux transport, signaling, membrane transport, toxin-antitoxin systems, and hypothetical proteins. SNP analysis identified 227 flipper rows corresponding to 194 unique genomic sites, including high-confidence mutations in stress-response, translation, autotransporter, and BapA-domain genes. Together, these results suggest that reversible antibiotic sensitivity is governed by a small, highly structured subset of lineage-specific genomic mutations rather than diffuse genome-wide variation, with indels acting as dominant structural drivers and SNPs contributing targeted regulatory or functional modulation. These findings advance computational genomics approaches for analyzing microbial evolution and establish a foundation for predictive modeling and genomic decision-support systems for antibiotic cycling strategies.**

## I. Introduction

The *Burkholderia cepacia* complex (Bcc) comprises a group of genetically related Gram-negative bacteria capable of establishing chronic lung infections in cystic fibrosis (CF) patients. Although Bcc prevalence is relatively low (approximately 3% of CF cases), these infections are strongly associated with increased morbidity, mortality, and the potentially fatal "cepacia syndrome" [1], [2]. Advances in CFTR-modulator therapy have improved patient survival [3], yet antimicrobial resistance remains a major barrier to effective long-term infection management. Bcc species readily evolve resistance through diverse mechanisms, including efflux pump regulation, stress-response mutations, alterations in cell permeability, and modification of antibiotic targets [4], [5].

In the experimental evolution dataset used for this study, *Burkholderia* strains were evolved under six antibiotics: chloramphenicol (CHL), ceftazidime (CAZ), levofloxacin (LVX), meropenem (MEM), minocycline (MIN), and trimethoprim-sulfamethoxazole (SXT). Across 279 evolved strains, 188 exhibited cross-resistance and 170 exhibited collateral sensitivity, demonstrating that resistance and sensitivity shifts occurred frequently across the lineage tree. This structure provides an opportunity to study whether specific genomic mutations repeatedly appear and disappear as strains transition between resistant and sensitive states.

A clinically relevant evolutionary response to antibiotic pressure is collateral sensitivity (CS), where acquisition of resistance to one antibiotic results in increased sensitivity to another [9]. Collateral sensitivity interactions may be reciprocal or asymmetrical. Reciprocal CS (RCS) pairs may enable long-term treatment strategies involving alternating antibiotics: as resistance to one drug emerges, sensitivity to its partner drug reappears, allowing clinicians to cycle treatments and suppress MDR evolution. Identifying the genetic underpinnings of CS and RCS is essential for translating this concept into clinical practice.

Experimental evolution studies of *Burkholderia multivorans* have produced hundreds of lineages with diverse resistance and sensitivity profiles (Fig. 1). Whole-genome sequencing reveals mutational patterns associated with CS, dominated by high-impact insertion–deletion (indel) events. However, manually interpreting lineage-specific mutation gain–loss patterns is not feasible at scale. A computational framework is needed to detect reversible allele states ("flippers") that track with antibiotic sensitivity transitions.

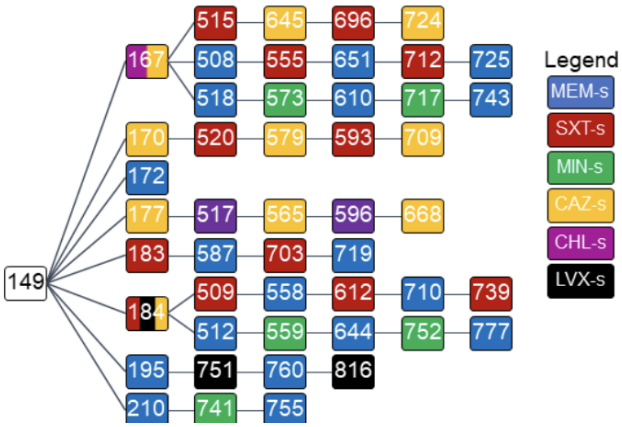


Fig. 1. Lineage tree of evolved *Burkholderia multivorans* strains derived from parental strain 149. Each node is labeled by strain ID and colored by the antibiotic to which collateral sensitivity was observed. Branching structure reflects sequential experimental evolution and sensitivity transitions.

### A. Research Questions

- Which genes exhibit reversible mutation states across antibiotic sensitivity transitions?
- Do perfect flippers correspond to known pathways in resistance, stress response, or cell-envelope regulation?
- Are indels disproportionately represented among CS-associated mutations?

## II. Related Work

Collateral sensitivity has been reported across multiple bacterial species and antibiotic classes, offering potential for evolution-informed therapy design [9]. Prior studies demonstrate that resistance pathways may incur tradeoffs, such as reduced efflux activity or altered metabolic burden, leading to increased susceptibility to unrelated drugs. In CF pathogens, including *Pseudomonas aeruginosa* and Bcc, repeated within-host evolution generates diverse mutational trajectories that shape antibiotic susceptibility [4], [5].

Genome-wide analyses show that indels often exert disproportionately strong biological effects compared to SNPs, disrupting reading frames and producing premature stop codons [10]. *Burkholderia* mutation-accumulation studies also highlight the roles of DNA repair pathways, stress regulators, porins, penicillin-binding proteins, virulence factors, and efflux transporters in resistance evolution [13]. Yet, computational approaches for integrating lineage structure with mutation dynamics remain underdeveloped.

Recent work in mutation spectrum analysis has utilized consensus-based calling, comparative genomics, and context-aware annotation pipelines [11], [12]. However, no existing framework is optimized specifically for detecting reversible mutation states across lineage trees—an essential requirement for identifying genetic drivers of CS and RCS.

## III. Methods

### A. Data Sources and Preprocessing

The dataset was derived from previously generated experimental evolution studies of *Burkholderia* lineages. Whole-genome sequencing data were obtained as BWA-aligned variant call datasets.

Variant calls were derived from VCF files and provided as filtered SNP and indel matrices:

- `burk_merged_filtered_well_snp.vcf`
- `burk_merged_filtered_well_indel.vcf`

Whole-genome sequencing was performed using 250 bp paired-end Illumina HiSeq 2500 reads. Reads were aligned to the reference genome using BWA, and pileup-derived files were processed to generate per-position read information across strains. SNP support files encoded forward and reverse read counts for each nucleotide base in the format `A, a, C, c, G, g, T, t`, while indel support files encoded read depth, indel count, and indel identity at each site.

These files represent mutation calls across all strains and evolutionary lineages relative to a reference genome. The flipper analysis focused on variant positions, while genome-wide support files such as `printBases`, `printIndel`, and BWA consensus outputs were used to calculate read-level allele frequencies and validate candidate sites across strains.

To ensure high-confidence mutation detection, a minimum read-support threshold of 75 was applied during preprocessing, consistent with stringent read-support filtering commonly applied in BWA-aligned variant datasets [14]. Variant tables were normalized into structured tabular formats, enabling downstream lineage-aware analysis.

### B. Pipeline Architecture

Two parallel computational pipelines were developed in Python to process SNP and indel datasets independently while maintaining consistent analytical structure. Each pipeline consisted of modular scripts designed for scalable genomic data processing.

The workflows included:

**1) Parsing and Normalization:** Variant tables were parsed to standardize column structure across strains, extract allele frequency metrics (e.g., *_pct columns), and remove incomplete or reference-only entries.

**2) Lineage Mapping:** A lineage dictionary was used to map strain identifiers to predefined evolutionary paths. This enabled grouping of strains into biologically meaningful trajectories reflecting antibiotic exposure histories.

**3) Flipper Detection:** Mutation "flippers" were identified as genomic sites exhibiting reversible allele states (reference or alternate) across lineage paths. Detection was performed using lineage-aware comparisons, where mutation presence and absence patterns were evaluated across sequential evolutionary steps.

**4) Annotation and Functional Mapping:** Variant positions were annotated using genomic context files, assigning gene

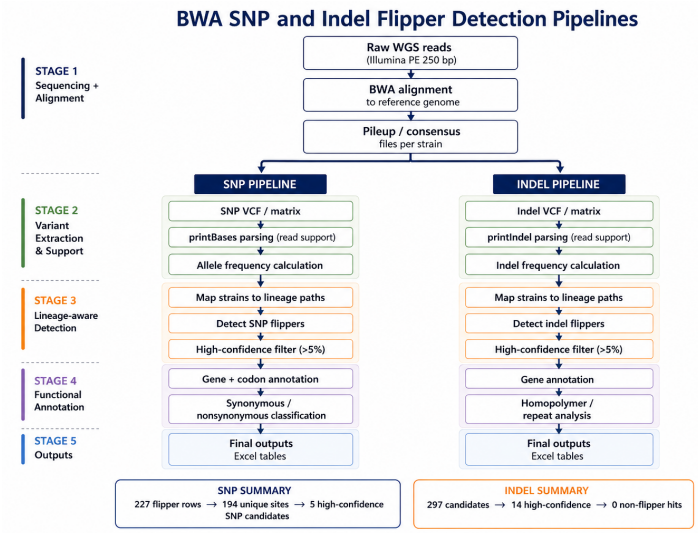


Fig. 2. Overview of the SNP and indel flipper detection pipelines. Raw sequencing reads are aligned using BWA, followed by variant extraction, lineage-aware flipper detection, functional annotation, and high-confidence filtering.

identifiers, locus tags, protein products, and coding information (codon and amino acid changes). Variants were classified as genic (EX) or intergenic (IG), with additional labeling for synonymous and nonsynonymous effects.

**5) Homopolymer and Sequence Context Analysis (Indels):** Indel variants were evaluated for sequence instability features, including homopolymer presence, motif structure, and repeat length. These features were used to distinguish biologically meaningful mutations from sequencing artifacts.

**6) Transition Metrics:** For each candidate site, transition counts and transition percentages were computed across strains, capturing the frequency of allele switching events within lineages.

**7) High-Frequency Filtering:** Candidate flipper sites were filtered based on allele frequency thresholds, retaining sites with at least one strain exhibiting greater than 5% variant frequency. This ensured that retained mutations had measurable read-level support.

**8) Output Generation:** Final outputs were exported as structured Excel workbooks, including:

- Flipper candidate tables
- High-frequency mutation subsets
- Genome-wide variant matrices
- Matched flipper-highfrequency datasets

### C. Implementation

All pipelines were implemented in Python using scalable data processing techniques. Pandas-based workflows enabled efficient handling of large genomic datasets, including files exceeding one million rows. Modular script design ensured reproducibility across SNP and indel analyses, while minimizing memory overhead during parsing and filtering operations.

## IV. Results

### A. Experimental Evolution and Mutation Landscape

The experimental evolution dataset contained 279 evolved strains derived from *Burkholderia* lineages exposed to CHL, CAZ, LVX, MEM, MIN, and SXT. Of these evolved strains, 188 exhibited cross-resistance and 170 exhibited collateral sensitivity. Whole-genome sequencing and mutation analysis were then used to identify genomic changes associated with these resistance-sensitivity transitions.

Initial analysis of the evolved lineage dataset indicated that mutation calls were strongly dominated by indels rather than SNPs. Because insertion–deletion events and single-nucleotide substitutions differ in frequency, biological impact, and downstream interpretation, separate SNP and indel workflows were developed to analyze structural mutation patterns independently from point mutation patterns.

### B. Indel Flipper Detection Identifies a Small Set of High-Confidence Mutations

Analysis of BWA-aligned indel datasets identified 297 candidate flipper rows, which were reduced to 14 high-confidence sites after applying an allele frequency threshold (¿5% in at least one strain). These indel flippers exhibited strong read-level support, with multiple sites reaching complete fixation (100% allele frequency) and high transition consistency across lineage paths.

All high-confidence indel flippers were strictly lineage-specific, with zero occurrences outside their evolutionary trajectories, indicating that these mutations represent localized adaptive responses rather than shared background variation.

### C. Functional Characterization of Indel Flippers

High-confidence indel flippers were enriched in genes associated with membrane transport, signaling, and stress response. Several key candidates include:

- **A8H40_RS09105 (nucleotidyl transferase / toxin-antitoxin system)** — A -1 bp deletion resulting in a frameshift and premature stop codon (100% frequency). This suggests disruption of toxin-antitoxin regulation, potentially altering stress survival mechanisms.
- **A8H40_RS24255 (efflux transporter outer membrane subunit)** — Multiple indel events (+1t, -1a) introducing frameshifts. Efflux systems are directly linked to antibiotic resistance, indicating a strong mechanistic connection to sensitivity switching.
- **A8H40_RS21255 (sensor histidine kinase)** — Frameshift mutation disrupting a signaling protein involved in environmental response, suggesting altered regulatory adaptation under antibiotic pressure.
- **A8H40_RS18565 (ABC transporter permease)** — Indel mutations affecting transport proteins that regulate substrate movement across the membrane.
- **A8H40_RS15555 (ComF family protein)** — DNA processing-related protein, indicating possible involvement in repair or recombination pathways.

TABLE I
REPRESENTATIVE HIGH-CONFIDENCE FLIPPER CANDIDATES

| Type | Locus | Product | Mutation Effect |
|---|---|---|---|
| Indel | A8H40_RS09105 | toxin-antitoxin protein | premature stop codon |
| Indel | A8H40_RS24255 | efflux outer membrane subunit | frameshift disruption |
| SNP | A8H40_RS05660 | (p)ppGpp synthetase/hydrolase | V→G substitution |
| SNP | A8H40_RS23195 | autotransporter-associated protein | A→V substitution |

- **A8H40_RS31205 / A8H40_RS24440 (hypothetical proteins)** — High-confidence indels with strong read support, representing potential novel contributors to resistance mechanisms.
- **Intergenic indels (e.g., position 1748809, 165538)** — Likely regulatory mutations affecting promoter or transcriptional control regions.

Many indel sites also occurred in or near homopolymer regions, suggesting sequence instability as a contributing factor to mutation formation.

### *D. SNP Flipper Detection Reveals Targeted Regulatory and Functional Changes*

The SNP pipeline identified 227 flipper rows corresponding to 194 unique genomic sites. After correcting parsing and lineage alignment, a refined subset of high-confidence SNP flippers was identified with strong allele frequency support and consistent transition behavior.

Key SNP flipper sites include:

- **A8H40_RS05660 ((p)ppGpp synthetase/hydrolase)** — Nonsynonymous mutation (V→G) with 100% frequency across multiple strains. This gene regulates the stringent response, a global stress adaptation pathway directly linked to antibiotic tolerance.
- **A8H40_RS08140 (threonine–tRNA ligase)** — Synonymous mutation with complete fixation, suggesting potential codon usage or translational efficiency effects.
- **A8H40_RS23195 (autotransporter-associated protein)** — Nonsynonymous mutation (A→V) affecting a membrane-associated virulence-related protein, indicating possible structural or functional changes impacting host interaction.
- **A8H40_RS21335 (BapA domain-containing protein)** — Synonymous mutation with high transition consistency (100%), suggesting subtle regulatory or expression-level effects.
- **Intergenic SNP (position 512984)** — High-frequency mutation likely affecting regulatory elements controlling nearby gene expression.

These SNPs were predominantly associated with regulatory, metabolic, and membrane-related functions, indicating a role in fine-scale adaptation rather than structural disruption.

### *E. Comparative Mutation Patterns Between Indels and SNPs*

Comparison of mutation classes revealed a clear distinction in biological impact. Indels, although less frequent, consistently exhibited higher allele frequency support, stronger transition consistency, and greater disruption of coding sequences through frameshift mutations and premature stop codons. In contrast, SNP flippers were more abundant but displayed greater variability and primarily resulted in amino acid substitutions or synonymous changes.

These results suggest that indels function as primary drivers of resistance-sensitivity transitions, while SNPs contribute secondary regulatory or modulatory effects within adaptive pathways.

## V. DISCUSSION

The results of this study demonstrate that reversible antibiotic sensitivity in *Burkholderia* is driven by a relatively small subset of genomic mutations rather than widespread genomic variation. The identification of a limited number of high-confidence flipper sites, particularly among indels, suggests that resistance-sensitivity transitions are governed by discrete and repeatable genetic events. These reversible mutation patterns provide a direct genomic correlate to collateral sensitivity, suggesting that flipper mutations may function as molecular switches that toggle between resistance and sensitivity states under changing antibiotic pressures.

This interpretation is consistent with the broader experimental evolution results from the Burkholderia study. The original mutation landscape was strongly indel-dominated, and many mutations occurred in genes connected to antibiotic response mechanisms, including efflux transporters, porins, penicillin-binding proteins, stress regulators, and cell-envelope functions [15]–[18]. These categories are biologically relevant because several antibiotics in this study act through membrane permeability, drug transport, cell-wall synthesis, or protein synthesis pathways. Therefore, reversible mutations affecting transport, signaling, or gene regulation provide plausible mechanistic links between genotype and collateral sensitivity phenotype.

Indel mutations emerged as the most biologically significant signals within the dataset. Their consistent high allele frequency support, strong transition behavior, and enrichment in coding regions indicate that they function as high-impact structural drivers of adaptation. The frequent introduction of frameshift mutations and premature stop codons suggests that these variants directly disrupt protein function, particularly in genes associated with membrane transport, signaling, and stress response pathways. These disruptions likely alter key cellular processes involved in antibiotic susceptibility.

In contrast, SNP flipper candidates exhibited greater variability and a broader distribution across genomic sites. However, refinement of the SNP pipeline revealed a subset of high-confidence SNPs with strong biological relevance. These mutations were enriched in genes involved in regulatory and metabolic processes, including global stress response and protein synthesis. Unlike indels, SNPs primarily resulted in amino acid substitutions or synonymous changes, suggesting

a role in fine-scale modulation of cellular function rather than large-scale structural disruption.

This distinction supports a hierarchical model of mutation impact, in which indels act as primary drivers of reversible phenotypic changes, while SNPs contribute secondary or modulatory effects within adaptive pathways. The coexistence of these mutation types indicates that reversible antibiotic sensitivity arises from both structural and regulatory layers of genomic adaptation.

The strict lineage specificity observed across all high-confidence flipper candidates further emphasizes the context-dependent nature of resistance evolution. These mutations were confined to specific evolutionary paths and were not observed in unrelated strains, indicating that adaptive responses are shaped by localized selection pressures rather than global population-level trends. This finding has important implications for antibiotic treatment strategies, suggesting that effective interventions must account for lineage-specific evolutionary dynamics.

From a computational perspective, this work highlights the importance of scalable bioinformatics pipelines in enabling discovery within large genomic datasets. Efficient processing of variant data, including millions of genomic positions across multiple strains, allowed for systematic identification of rare but biologically meaningful mutation patterns. The modular pipeline architecture supports reproducibility and can be extended to additional datasets or microbial systems.

Looking forward, these pipelines provide a foundation for developing predictive models that link mutation patterns to antibiotic response phenotypes. Integration of these models into full-stack genomic analytics platforms could enable real-time visualization, lineage tracking, and data-driven decision-making in clinical contexts. Such systems have the potential to support antibiotic cycling strategies and improve treatment outcomes by leveraging evolutionary insights at the genomic level.

## VI. Conclusion

This study demonstrates that reversible antibiotic sensitivity in *Burkholderia* is governed by a small, well-defined subset of genomic mutations rather than widespread genomic variation. By developing scalable, lineage-aware bioinformatics pipelines, this work enabled systematic identification of mutation "flippers" that track with resistance-sensitivity transitions across experimentally evolved lineages.

Indel mutations emerged as the most consistent and biologically impactful signals, exhibiting strong read-level support, high transition consistency, and frequent disruption of coding sequences through frameshifts and premature stop codons. These mutations were enriched in genes associated with membrane transport, regulatory signaling, and stress response pathways, indicating direct functional consequences on cellular mechanisms relevant to antibiotic susceptibility. In contrast, SNP flippers, while more abundant, demonstrated greater variability and primarily contributed targeted, fine-scale modifications within key regulatory and metabolic genes. Together, these findings support a hierarchical model of mutation impact, where indels act as primary drivers of reversible phenotypic change and SNPs provide secondary or modulatory effects.

A defining characteristic of all high-confidence flipper candidates was strict lineage specificity, with no observed recurrence outside their respective evolutionary paths. This highlights that resistance evolution in *Burkholderia* is highly context-dependent and shaped by localized selection pressures rather than global mutation trends. As a result, effective antibiotic strategies must account for lineage-specific mutation dynamics rather than relying solely on population-level assumptions.

Beyond the biological insights, this work establishes a robust computational framework for analyzing complex microbial evolution at scale. The pipelines developed in this study efficiently process large variant datasets, integrate lineage structure, and extract biologically meaningful mutation patterns in a reproducible manner. This approach enables systematic exploration of genotype-phenotype relationships that are otherwise difficult to resolve through manual analysis.

Looking forward, these pipelines can serve as the foundation for predictive modeling of antibiotic response. By integrating mutation detection with machine learning approaches, it becomes possible to learn relationships between genomic variation and phenotypic outcomes. These models can be further translated into full-stack genomic analytics platforms that combine mutation tracking, lineage visualization, and real-time data interpretation. Such systems have the potential to support data-driven antibiotic cycling strategies and improve clinical decision-making by leveraging evolutionary genomics.

Overall, this work advances both the computational and biological understanding of collateral sensitivity and provides a scalable framework for translating genomic data into actionable insights for antibiotic treatment optimization.

## References


[1] B. Kerem *et al.*, "Identification of the cystic fibrosis gene: genetic analysis," *Science*, vol. 245, pp. 1073–1080, 1989.

[2] K. A. Spoonhower and P. B. Davis, "Epidemiology of cystic fibrosis," *Clin. Chest Med.*, vol. 37, no. 1, pp. 1–10, 2016.

[3] Cystic Fibrosis Foundation, "Patient Registry Annual Data Report," 2021.

[4] T. J. Kidd *et al.*, "Defining antimicrobial resistance in cystic fibrosis," *J. Cystic Fibrosis*, vol. 17, pp. 696–704, 2018.

[5] M. Starkey *et al.*, "Pseudomonas aeruginosa rugose small-colony variants have adaptations that likely promote persistence in the cystic fibrosis lung," *J. Bacteriol.*, vol. 191, pp. 3492–3503, 2009.

[6] W. J. Welch, "Role of quality control pathways in human diseases involving protein misfolding," *Semin. Cell Dev. Biol.*, vol. 15, pp. 31–38, 2004.

[7] G. S. Sawicki *et al.*, "Sustained Benefit from ivacaftor demonstrated by combining clinical trial and cystic fibrosis patient registry data," *Am. J. Respir. Crit. Care Med.*, vol. 192, pp. 836–842, 2015.

[8] F. Armougom *et al.*, "Microbial diversity in the sputum of a cystic fibrosis patient studied with 16S rDNA pyrosequencing," *Eur. J. Clin. Microbiol. Infect. Dis.*, vol. 28, pp. 1151–1154, 2009.

[9] A. Colclough *et al.*, "Patterns of cross-resistance and collateral sensitivity between clinical antibiotics and natural antimicrobials." *Evol. Appl.*, vol. 12, pp. 878–887, 2019.

[10] W. Sung *et al.*, "Evolution of the Insertion-Deletion Mutation Rate Across the Tree of Life," *G3*, vol. 6, pp. 2583–2591, 2016.

[11] M. Lynch *et al.*, "A genome-wide view of the spectrum of spontaneous mutations in yeast," *PNAS*, vol. 105, pp. 9272–9277, 2008.
[12] S. Ossowski *et al.*, "The rate and molecular spectrum of spontaneous mutations in *Arabidopsis thaliana*," *Science*, vol. 327, pp. 92–94, 2010.
[13] K. A. Rhodes and H. P. Schweizer, "Antibiotic resistance in *Burkholderia*," *Drug Resist. Updat.*, vol. 28, pp. 82–90, 2016.
[14] H. Li and R. Durbin, "Fast and accurate short read alignment with Burrows-Wheeler transform," *Bioinformatics*, vol. 25, no. 14, pp. 1754–1760, 2009.
[15] A. Asadi *et al.*, "Minocycline, focus on mechanisms of resistance, antibacterial activity, and clinical effectiveness: Back to the future," *Journal of Global Antimicrobial Resistance*, vol. 22, pp. 161–174, 2020.
[16] G. A. Jacoby, "Mechanisms of resistance to quinolones," *Clinical Infectious Diseases*, vol. 41, no. Supplement_2, pp. S120–S126, 2005.
[17] U. Choi and C.-R. Lee, "Distinct roles of outer membrane porins in antibiotic resistance and membrane integrity in *Escherichia coli*," *Frontiers in Microbiology*, vol. 10, 2019.
[18] C. J. Stubenrauch *et al.*, "A noncanonical chaperone interacts with drug efflux pumps during their assembly into bacterial outer membranes," *PLOS Biology*, vol. 20, no. 1, p. e3001523, 2022.
[19] L. E. Williams and J. J. Wernegreen, "Sequence context of indel mutations and their effect on protein evolution in a bacterial endosymbiont," *Genome Biology and Evolution*, vol. 5, no. 3, pp. 599–605, 2013.
[20] E. M. Windels *et al.*, "Bacterial persistence promotes the evolution of antibiotic resistance by increasing survival and mutation rates," *ISME Journal*, vol. 13, no. 5, pp. 1239–1251, 2019.